

[DC] Virtual Reality and Beyond: Exploring the Design and User Experience of Augmented Social Touch in VR

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ABSTRACT

This paper highlights the potential of exploring mediated social touch in virtual reality beyond the recreation of face-to-face interactions, motivating my proposed research structure to identify key design parameters of *augmented social touch* and to investigate the resulting immersive, social, and emotional user experience.

Index Terms: Augmented social touch, mediated social touch, MST, social augmentations, virtual reality, VR, social VR

1 INTRODUCTION

Social touch is a fundamental human need and constitutes the most significant nonverbal channel for emotional communication [18], which is why mediated communication could greatly benefit from integrating touch. The common approach for this integration is to use haptic feedback devices, based on vibration motors, electrical pulses, or ultrasound [3], usually with the goal of realistically recreating the sensory experience of real touch at some point [10]. This focus on virtually recreating touch is understandable and backed by user demand [15], but there are also works highlighting the need “to rethink and reimagine social touch through a deeper engagement with the social and sensory aspects of touch” [11, p. 1].

One approach for this reimagination lies in exploring the *potential of digital technologies to deliberately deviate from reality*. This perspective builds on work that emphasizes the need to investigate the unique characteristics of an interactive medium to identify novel forms of communication [8] and to understand how people use and experience them during their establishment [6]. Examples of this include concepts such as *transformed social interactions* [2], i.e., the targeted decoupling of displayed behavior from actual behavior, and *social augmentations* [12], i.e., enhancing virtual communication beyond what is possible face-to-face. Both aim to change or enhance the social experience in virtual environments compared to the real world, rather than merely recreating reality.

2 THE POTENTIAL OF AUGMENTED SOCIAL TOUCH

Social augmentations have been explored in the form of decoupling user and avatar behaviors like gaze [2], translating social phenomena such as interpersonal distance [13] into visual artifacts (e.g., by coloring avatars based on proximity), or adding audio-visual stimuli to nonverbal interactions like mediated social touch (MST) [4, 17]. Since social touch plays a central role in promoting interpersonal intimacy [7], virtual reality (VR) as a medium has great potential to enable authentic and meaningful social experiences at a distance. MST can trigger affective [5, 16] and physiological [5] reactions depending on the body regions involved, the gender constellation [5, 16] and sexuality of the users [5], and the touch direction [16]. Additional haptic feedback stimuli appear to trigger more intense emotional and social reactions [14].

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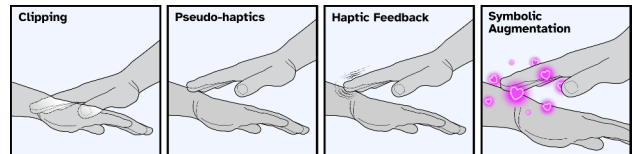


Figure 1: Comparison of common presentations of mediated social touch in VR with visually augmented social touch.

Even if not all users would describe an audio-visually presented social touch in VR as similar to an actual physical touch, they still report feeling heat, pressure, and a tingling [1] or tactile sensation [4]. For *augmented social touch*, there are also first indications for the effect of specific design parameters, such as warm colors evoking temperature sensations and a feeling of closeness [9], while additional symbolic effects, like heart-shaped patterns, are used to enhance intimate touch interactions [19]. Nevertheless, there has not yet been a systematic investigation of the user experience (UX) of *augmented social touch* in VR. This represents a significant research gap regarding the socio-emotional relevance of interpersonal touch [18] and the anticipated significance of technologically mediated social interactions [11, 14].

3 PROPOSED PHD RESEARCH

To bridge the research gap outlined above, I plan to develop a **framework** structuring MST in VR, based on the *kind of touch* itself (e.g., a hug), the *social context* of the touch (e.g., a friendly relationship), and the *presentation* of the touch (e.g., visually augmented). The framework will encompass an **augmentation design classification** (Fig. 2) building upon [17] and [12]. With this, I intend to contribute to a shared vocabulary for *augmented social touch*, enabling comparisons across different findings within the research area and providing a structure for investigating effects among the framework’s components.

My plan is to combine existing findings on MST and unmediated social touch with an **online survey** of social VR users, to integrate the perspective of potential users of *augmented social touch*, for exploring which contextual factors and social norms can be translated from unmediated touch and which may be specific to VR.

The augmentation design classification will include *visual*, *auditory*, and *haptic* design parameters (e.g., color or vibration intensity), derived from a literature review and mixed-methods **lab studies** focusing on augmentations using a single sensory channel (e.g., visual augmentations). Through these studies, I intend to identify the key design parameters and how they affect the UX. For this, the augmented touch interactions will be compared to established presentations of MST in VR like *clipping*, i.e., the absence of collisions between avatars, or *pseudo-haptics*, i.e., the illusion of touch, created by simulating collisions between avatars and allowing a certain deviation between the rendered hand position and the user’s real hand position (see Fig. 1).

The final framework and augmentation design classification will then be utilized to study multi-modal augmentations (combining visual, auditory, and haptic stimuli) using an interactive *augmen-*

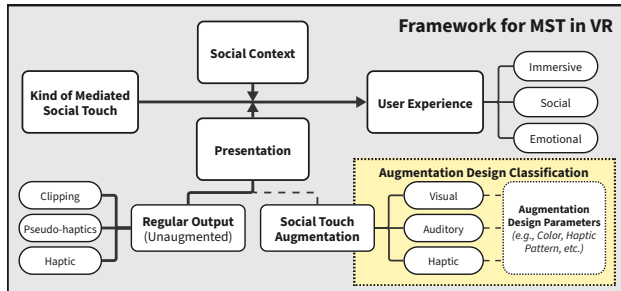


Figure 2: Overview of the proposed research framework and augmentation design classification for VR-mediated social touch.

tation editor in VR, which allows users to design and apply their own social touch augmentations. The editor will be part of an **exploratory lab study**, in which participants are asked to create augmentations for a given communicative goal or experience (e.g., an augmentation that feels pleasant). After designing the augmentation, participants will specify on a set of scales which communicative intent they associate with their design. Based on the data, I plan to explore overarching trends and correlations between specific experiential outcomes (e.g., social presence) and augmentation design parameters (e.g., warm colors).

Concluding this user-centered approach, a final set of augmentations will be derived and presented to novel users in a **confirmatory lab study** to assess whether the expected experience is actually evoked during augmented social touch. The expected contribution is a set of validated guidelines to inform the targeted design of social touch augmentations for specific UX outcomes.

4 CONCLUSION

To summarize, I see a significant potential for MST research to focus on the unique characteristics of VR, exploring emerging forms of communication beyond the replication of face-to-face interactions. Therefore, I intend to focus my PhD research on *augmented social touch* in VR, the contextual factors of using it, and how it can be designed to change or enhance the immersive, social, and emotional UX. Currently, I am exploring visual augmentations while working towards the technical implementation for the proposed lab studies. To strengthen my research approach, I would like to discuss the following topics during the doctoral consortium: (A) *What is an adequate granularity for the MST framework and augmentation design classification?* (B) *How can a touch experience questionnaire be designed to allow comparison between real, mediated, and augmented social touch, e.g., where would a face-to-face interaction be placed on the scale of such a questionnaire?* (C) *Which physiological measurements should be integrated into the study setup to infer aspects of the UX (e.g., emotional arousal) without relying exclusively on self-assessment?*

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